

(-) 分布函数: $F_X(x) = P\{X \leq x\}$				
(二) 各种分布:				
分布名称	写作	密度函数	EX	DX
二项分布	$X \sim B(n, p)$	$P(X=a) = C_n^a \cdot p^a \cdot (1-p)^{n-a}$	np	$np(1-p)$
几何分布	$X \sim g(p)$	$P(X=a) = p \cdot (1-p)^{a-1}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
泊松分布	$X \sim P(\lambda)$	$P(X=a) = \frac{\lambda^a \cdot e^{-\lambda}}{a!}$	λ	λ
均匀分布	$X \sim U[a, b]$	$p_X(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & \text{其他} \end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
指数分布	$X \sim E(\lambda)$	$p_X(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & \text{其他} \end{cases}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
正态分布	$X \sim N(\mu, \sigma^2)$	$p_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2

(三). 期望 $E(g(x)) = \int_{-\infty}^{+\infty} g(x) p_X(x) dx$ $E(aX+bY) = aEX + bEY$	
若 X, Y 独立, $E(X \cdot Y) = EX \cdot EY$	
方差 $D(X) = EX^2 - (EX)^2$ $D(aX+b) = a^2 D(X)$	
$D(X \pm Y) = D(X) + D(Y) \pm 2cov(X, Y)$	
协方差 $cov(X, Y) = EXY - EX \cdot EY = E[(X-EX)(Y-EY)]$	
$cov(aX+c, bY+d) = ab cov(X, Y)$	
$cov(X_1+X_2, Y) = cov(X_1, Y) + cov(X_2, Y)$	
相关系数 $\rho_{XY} = \frac{cov(X, Y)}{\sqrt{DX \cdot DY}}$	

(四). 切比雪夫不等式: $\forall$ 随机变量 $X, \varepsilon > 0, P(X-EX \geq \varepsilon) \leq \frac{DX}{\varepsilon^2}$	
中心极限定理: $X_1 \sim X_n$ 独立同分布于 $X, EX = \mu, DX = \sigma^2,$ 则 $\sum_{i=1}^n X_i \sim N(n\mu, n\sigma^2)$, $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$	

(五). 常用统计量: (设 $(X_1, X_2, \dots, X_n) \sim X$)	
样本均值 $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$	
样本方差 $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$	
样本二阶中心矩 $S^{*2} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$	
结论: 若 $EX = \mu, DX = \sigma^2$, 则 $EX_i = \mu$ $DX_i = \sigma^2$	
$EX = \mu$ $D\bar{X} = \frac{\sigma^2}{n}$	
$ES^2 = \sigma^2$	
(六): χ^2 分布: $(X_1, X_2, \dots, X_n) \sim N(0, 1)$ 则 $\sum_{i=1}^n X_i^2 \sim \chi^2(n)$	
$E\chi^2(n) = n$ $D\chi^2(n) = 2n$	
t 分布: $X \sim N(0, 1)$; $Y \sim \chi^2(n)$, 则 $\frac{X}{\sqrt{Y/n}} \sim t(n)$	
F 分布: $X \sim \chi^2(n_1)$; $Y \sim \chi^2(n_2)$, 则 $\frac{X/n_1}{Y/n_2} \sim F(n_1, n_2)$	

(七): 上 α 分位点: $U_{1-\alpha} = -U_\alpha$ $t_{1-\alpha}(n) = -t_\alpha(n)$ $F_{1-\alpha}(n_1, n_2) = \frac{1}{F_\alpha(n_2, n_1)}$	
(八): 正态总体: $(X_1, X_2, \dots, X_n) \sim N(\mu, \sigma^2)$, 则有:	
$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ $\frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$ $\frac{\bar{X}-\mu}{s/\sqrt{n}} \sim t(n-1)$	
(九). 置信区间: 已知 $(X_1, X_2, \dots, X_n) \sim N(\mu, \sigma^2)$ 。置信度为 $1-\alpha$ 。	
σ^2 已知, 估计 μ : $[\bar{X} - \frac{\sigma}{\sqrt{n}} U_{\frac{\alpha}{2}}, \bar{X} + \frac{\sigma}{\sqrt{n}} U_{\frac{\alpha}{2}}]$	
σ^2 未知, 估计 μ : $[\bar{X} - \frac{s}{\sqrt{n}} t_{\frac{\alpha}{2}}(n-1), \bar{X} + \frac{s}{\sqrt{n}} t_{\frac{\alpha}{2}}(n-1)]$	
μ 未知, 估计 σ^2 : $[\frac{(n-1)s^2}{\chi_{\frac{\alpha}{2}}^2(n-1)}, \frac{(n-1)s^2}{\chi_{1-\frac{\alpha}{2}}^2(n-1)}]$	
σ_1^2, σ_2^2 已知, 估计 $\mu_1 - \mu_2$: $[\bar{X} - \bar{Y} - \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} U_{\frac{\alpha}{2}}, \bar{X} - \bar{Y} + \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} U_{\frac{\alpha}{2}}]$	
$\sigma_1^2 = \sigma_2^2 = \sigma^2$ 未知, 估计 $\mu_1 - \mu_2$: $\frac{(\bar{X}-\bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{n_1 n_2 (n_1 + n_2 - 2)}{n_1 + n_2}} \cdot \frac{s}{\sqrt{(n_1-1)s^2 + (n_2-1)s^2}}} \sim t(n_1 + n_2 - 2)$ 现推	
μ_1, μ_2 未知, 估计 $\frac{\sigma_1^2}{\sigma_2^2}$: $[\frac{S_1^2}{S_2^2} \cdot \frac{1}{F_{\frac{\alpha}{2}}(n_1-1, n_2-1)}, \frac{S_1^2}{S_2^2} \cdot \frac{1}{F_{1-\frac{\alpha}{2}}(n_1-1, n_2-1)}]$	

(十). 假设检验: 设显著性水平为 $\alpha, (X_1, X_2, \dots, X_n) \sim N(\mu, \sigma^2)$	
① $H_0: \mu = \mu_0, \sigma^2$ 已知: $U = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$	$W = \{ U \geq U_{\frac{\alpha}{2}} \}$
② $H_0: \mu \geq \mu_0, \sigma^2$ 已知: $U = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$	$W = \{ U \leq -U_\alpha \}$
③ $H_0: \mu = \mu_0, \sigma^2$ 未知: $T = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$	$W = \{ T \geq t_{\frac{\alpha}{2}}(n-1) \}$
④ $H_0: \sigma^2 = \sigma_0^2, \mu$ 未知: $\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$	$W = \{ \chi^2 \geq \chi_{\frac{\alpha}{2}}^2(n-1) \} \cup \{ \chi^2 \leq \chi_{1-\frac{\alpha}{2}}^2(n-1) \}$
⑤ $H_0: \mu_1 = \mu_2, \sigma_1^2, \sigma_2^2$ 已知: $U = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$	$W = \{ U \geq U_{\frac{\alpha}{2}} \}$
⑥ $H_0: \mu_1 = \mu_2, \sigma_1^2 = \sigma_2^2 = \sigma^2$ 未知: $T = \frac{\sqrt{\frac{n_1 n_2 (n_1 + n_2 - 2)}{n_1 + n_2}} \cdot \frac{\bar{X} - \bar{Y}}{\sqrt{(n_1-1)s_1^2 + (n_2-1)s_2^2}}}{\sqrt{(n_1-1)s_1^2 + (n_2-1)s_2^2}}$	$W = \{ T \geq t_{\frac{\alpha}{2}}(n_1 + n_2 - 2) \}$
⑦ $H_0: \sigma_1^2 = \sigma_2^2, \mu_1, \mu_2$ 未知: $F = \frac{S_1^2}{S_2^2}$	$W = \{ F \leq F_{1-\frac{\alpha}{2}}(n_1-1, n_2-1) \} \cup \{ F \geq F_{\frac{\alpha}{2}}(n_1-1, n_2-1) \}$